SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5116
COURSE NAME: Cryptography and Network Security

192373051

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage: **40%**

QUESTIONS:

Total Marks: 50

Annexure A

Analytical Problem Solving

Code of Conduct:

I G. Vignesh *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: G. Vignesh.

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n ϕ(n)\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i. Calculate total encryption time for each algorithm
ii. Analyze trade-offs between security and performance
iii. Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison
Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganize d steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

2. Block cipher Mode Evaluation.

Given

- Plaintext blocks: $P = [1010, 1010, 1111, 1010]$
- key : $K = 1100$
- Initialization Vector (IV) : 0000

* 4 bit Encryption function.

$$E_K(X) = X \oplus K$$

a) Encryption using ECB and CBC modes.

i) ECB (Electronic Codebook)

Formula : $C_i = P_i \oplus K$

Block-1 : $C_1 = 1010 \oplus 1100 = 0110$

Block-2 : $C_2 = 1010 \oplus 1100 = 0110$

Block-3 : $C_3 = 1111 \oplus 1100 = 0011$

Block-4 : $C_4 = 1010 \oplus 1100 = 0110$

$C_{ECB} = [0110, 0110, 0011, 0110]$

ii) CBC (Cipher block chaining)

Formula : $C_i = (P_i \oplus C_{i-1}) \oplus K$

IV = 0000

- $C_1 = (1010 \oplus 0000) \oplus 1100 = 0110$
- $C_2 = (1010 \oplus 0110) \oplus 1100 = 0000$
- $C_3 = (1111 \oplus 0000) \oplus 1100 = 0011$
- $C_4 = (1010 \oplus 0011) \oplus 1100 = 0101$

$C_{CBC} = [0110, 0000, 0011, 0101]$

b) Compare ciphertext outputs :

ECB	CBC
0110, 0110, 0011, 0110	0110, 0000, 0011, 0101
Encrypts blocks independently	Links blocks using previous ciphertext.
Repeated plaintext gives repeated Cipher text	Repeated plaintext gives different Ciphertext
No IV required	Requires an IV.
Pattern Visibility more	Better pattern concealment

c) Which model hides patterns better ?

* CBC hides pattern better than ECB because :

d) Analyze Computational cost for both Systems.

*RSA :

1) Uses large-integer modular exponentiation.

2) Requires more computation for private key operations

3). May take more processing time on low-power devices

4). Larger key increases computational requirements

5) Private-key operations are Very expensive

*ECC :

1) It uses mathematical operations on elliptic Curve points

2) It provides comparable Security with smaller key

3) It is suitable for microcontrollers and low-power devices

4) It can reduce processing time when implemen-ted efficiently.

5) Actual performance depends on the algorithm implementation and hardware.

1. Each block depends on the previous ciphertext block

Identical plaintext blocks can produce different ciphertext

The IV influences the first block

Repeated patterns are less visible to an attacker

Conclusion: CBC provides better pattern concealment than ECB.

5. ECC Vs RSA Comparision.

a) Introduction.

RSA and ECC are public-key cryptographic algorithms used for secure communication, key exchange and digital signatures.

Given:

a) RSA key size = 1024 bits

b) ECC key size = 160 bits

c) Device = resource-limited IoT device.

e) Estimate which Consumes less power & memory

* RSA

→ Requires more memory to store keys and relat
-ed data

→ Usually produces larger Signatures.

→ Larger message can increase communication overhead.

→ Private-key operations may consume more energy.

* ECC

→ Requires less memory of smaller keys

→ produces smaller signatures

→ Reduces data transmission overhead

→ Helps extend battery life in resource-constrained devices.

Result :-

ECC consumes less memory and power

f). Which is better for IoT ?

ECC is generally the better choice because :

It uses smaller keys

It requires less memory.

It reduces communication overhead

It is often more energy-efficient

Conclusion :-

Ecc is generally preffered over RSA for resource limited IoT devices because it provides comparable security with smaller keys and lower storage.